

Graphing Linear Equations

Method I: Finding Two Random Points

① $x + 2 = y$

x	y
0	2
1	3

1st Point: $x = 0$
 $\Downarrow$
 $0 + 2 = y$
 $2 = y$
 $(0, 2)$

2nd Point: $x = 1$
 $\Downarrow$
 $1 + 2 = y$
 $3 = y$
 $(1, 3)$

② $x = \frac{y}{2} - 1$

x	y
-1	0
0	2

1st Point: $y = 0$
 $\Downarrow$
 $x = \frac{0}{2} - 1$
 $x = 0 - 1$
 $x = -1$
 $(-1, 0)$

2nd Point: $y = 2$
 $\Downarrow$
 $x = \frac{2}{2} - 1$
 $x = 1 - 1$
 $x = 0$
 $(0, 2)$

③ $y = -5 - 3x$

x	y
-1	-2
-2	1

1st Point: $x = -1$
 $\Downarrow$
 $y = -5 - 3(-1)$
 $y = -5 + 3$
 $y = -2$
 $(-1, -2)$

2nd Point: $x = -2$
 $\Downarrow$
 $y = -5 - 3(-2)$
 $y = -5 + 6$
 $y = 1$
 $(-2, 1)$

④ $x = y - 1$

x	y
-1	0
0	1

1st Point: $y = 0$
 $\Downarrow$
 $x = 0 - 1$
 $x = -1$
 $(-1, 0)$

2nd Point: $y = 1$
 $\Downarrow$
 $x = 1 - 1$
 $x = 0$
 $(0, 1)$

⑤ $-2x + 4 = y$

x	y
1	2
2	0

1st Point: $x = 1$
 $\Downarrow$
 $-2(1) + 4 = y$
 $-2 + 4 = y$
 $2 = y$
 $(1, 2)$

2nd Point: $x = 2$
 $\Downarrow$
 $-2(2) + 4 = y$
 $-4 + 4 = y$
 $0 = y$
 $(2, 0)$

⑥ $y = -\frac{x}{3} + 2$

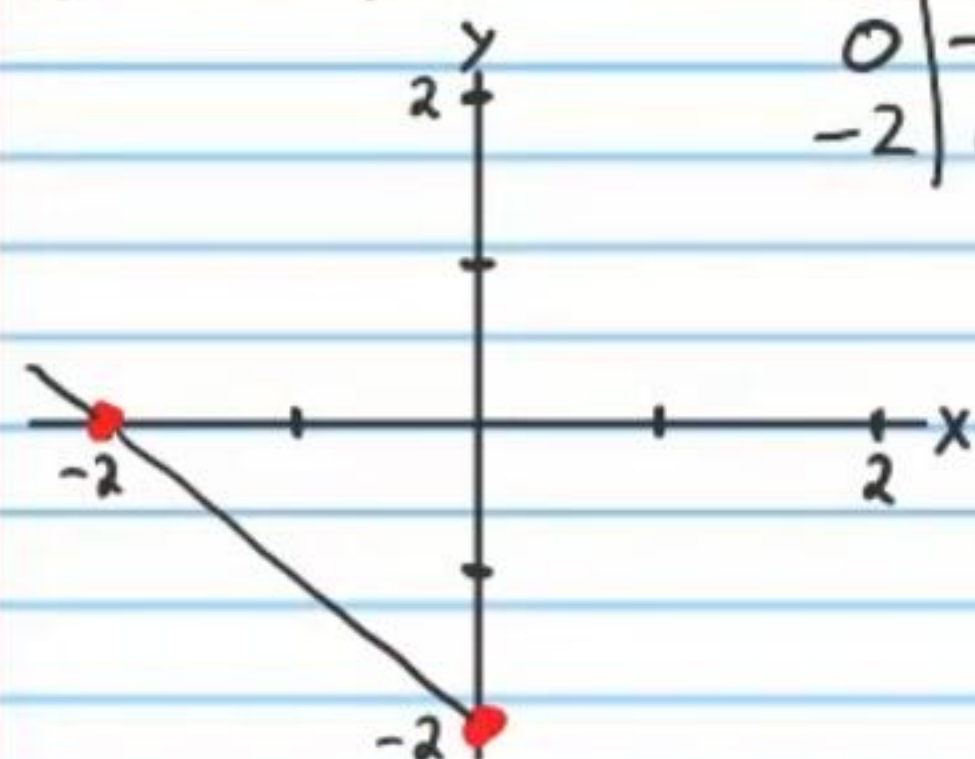
x	y
3	1
0	2

1st Point: $x = 0$
 $\Downarrow$
 $y = -\frac{0}{3} + 2$
 $y = 0 + 2$
 $y = 2$
 $(0, 2)$

2nd Point: $x = 3$
 $\Downarrow$
 $y = -\frac{3}{3} + 2$
 $y = -1 + 2$
 $y = 1$
 $(3, 1)$

⑦ $X + Y = -2$

X	Y
0	-2
-2	0



1st Point

$X = 0$



$0 + Y = -2$

$Y = -2$

$(0, -2)$

2nd Point

$Y = 0$

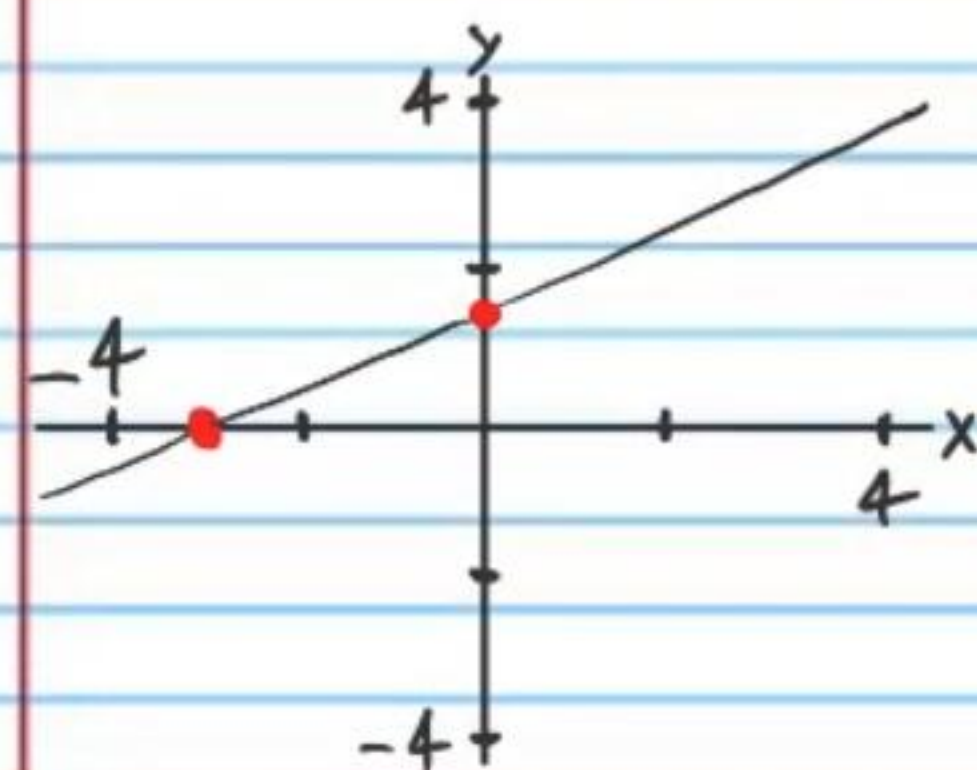


$X + 0 = -2$

$X = -2$

$(-2, 0)$

⑧ $2Y - X = 3$



1st Point

$Y = 0$



$2(0) - X = 3$

$0 - X = 3$

$-X = 3$

$-1X = 3$

$X = -3$

$(-3, 0)$

2nd Point

$X = 0$



$2Y - 0 = 3$

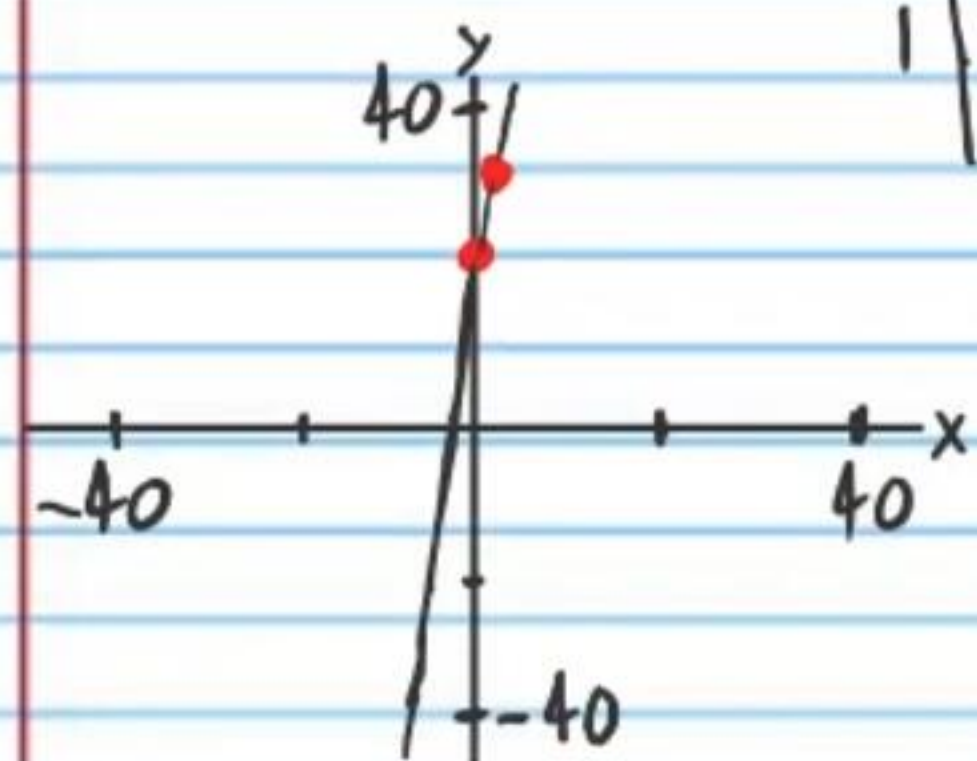
$2Y = 3$

$Y = 3/2$

$(0, 3/2)$

⑨ $\frac{Y}{5} = 2X + 4$

X	Y
0	20
1	30



1st Point

$X = 0$



$\frac{Y}{5} = 2(0) + 4$

$\frac{Y}{5} = 0 + 4$

$\frac{Y}{5} = 4$

$Y = 20$

2nd Point

$X = 1$



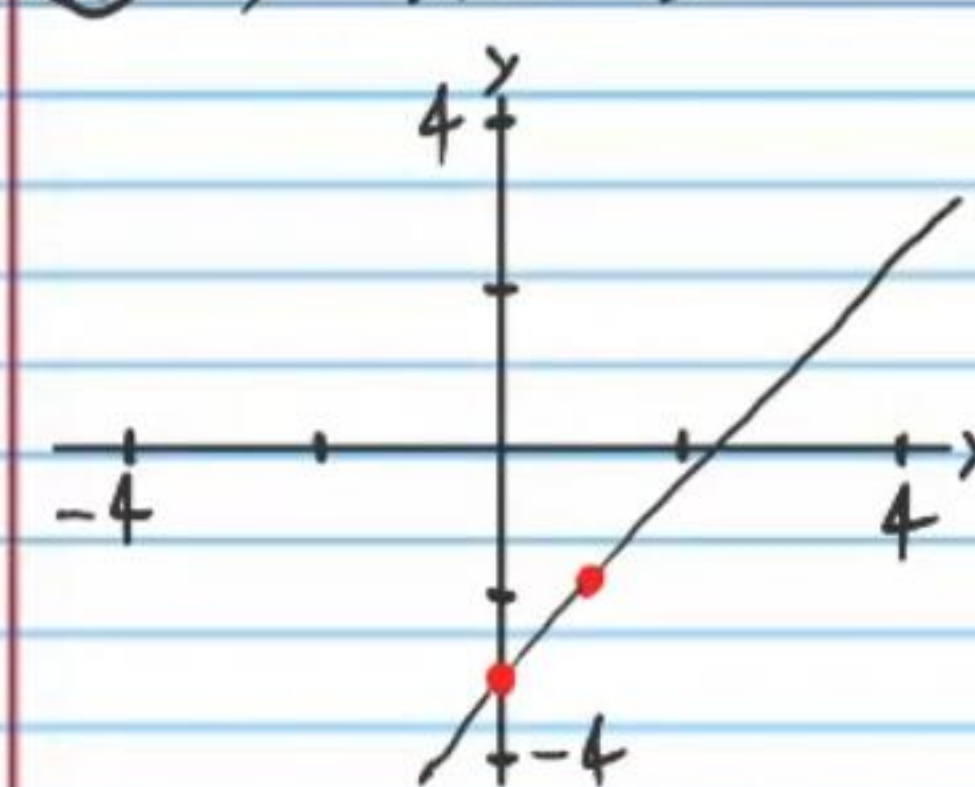
$\frac{Y}{5} = 2(1) + 4$

$\frac{Y}{5} = 2 + 4$

$\frac{Y}{5} = 6$

$Y = 30$

⑩ $Y - X = -3$



1st Point

$X = 0$



$Y - 0 = -3$

$Y = -3$

$(0, -3)$

2nd Point

$X = 1$

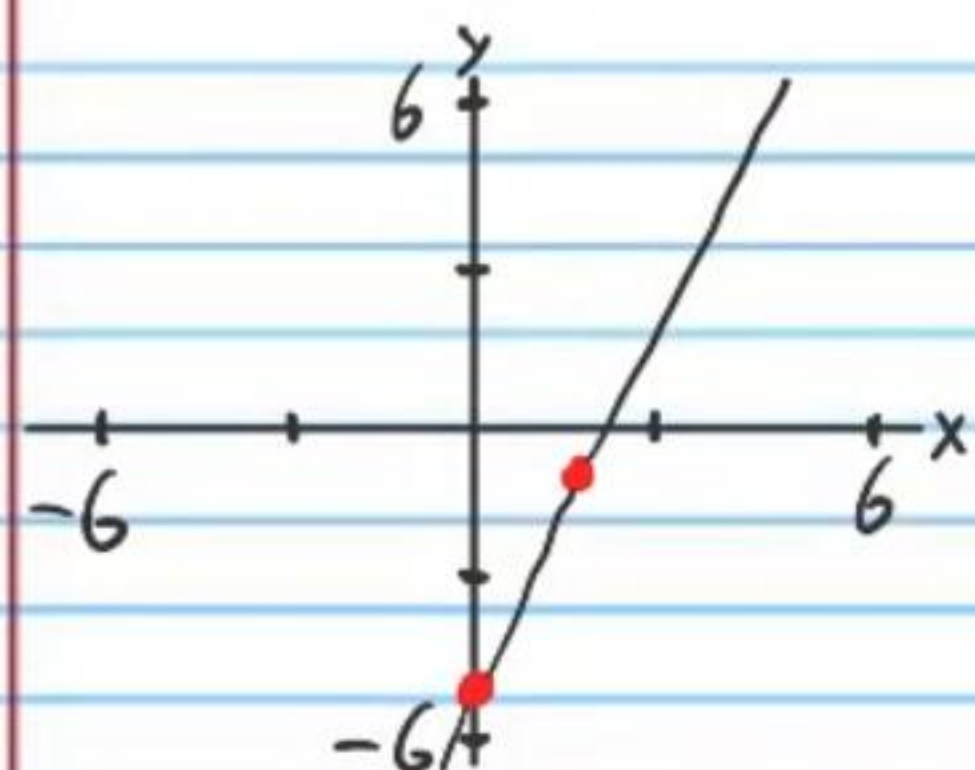


$Y - 1 = -3$

$Y = -2$

$(1, -2)$

⑪ $5 = -Y + 2X$



1st Point

$X = 0$



$5 = -Y + 2(0)$

$5 = -Y + 0$

$5 = -Y$

$-5 = Y$

$(0, -5)$

2nd Point

$X = 2$



$5 = -Y + 2(2)$

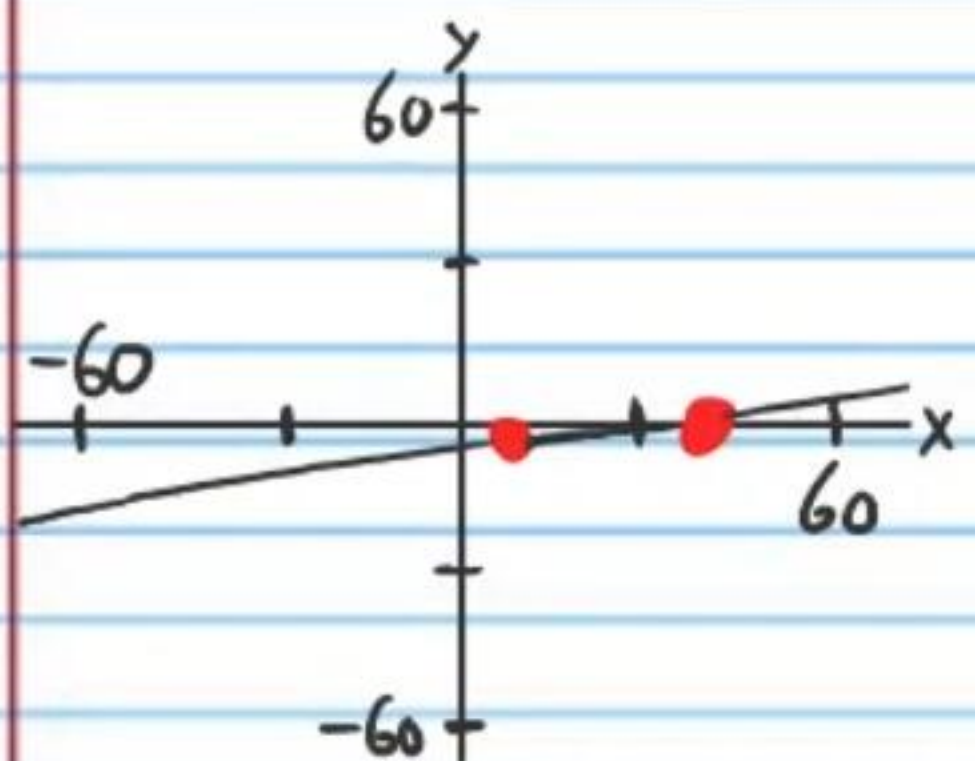
$5 = -Y + 4$

$1 = -Y$

$-1 = Y$

$(2, -1)$

⑫ $\frac{X}{4} - 10 = 4Y$



1st Point

$Y = 0$



$\frac{X}{4} - 10 = 4(0)$

$\frac{X}{4} - 10 = 0$

$X = 40$

$(40, 0)$

2nd Point

$X = 8$



$\frac{8}{4} - 10 = 4Y$

$2 - 10 = 4Y$

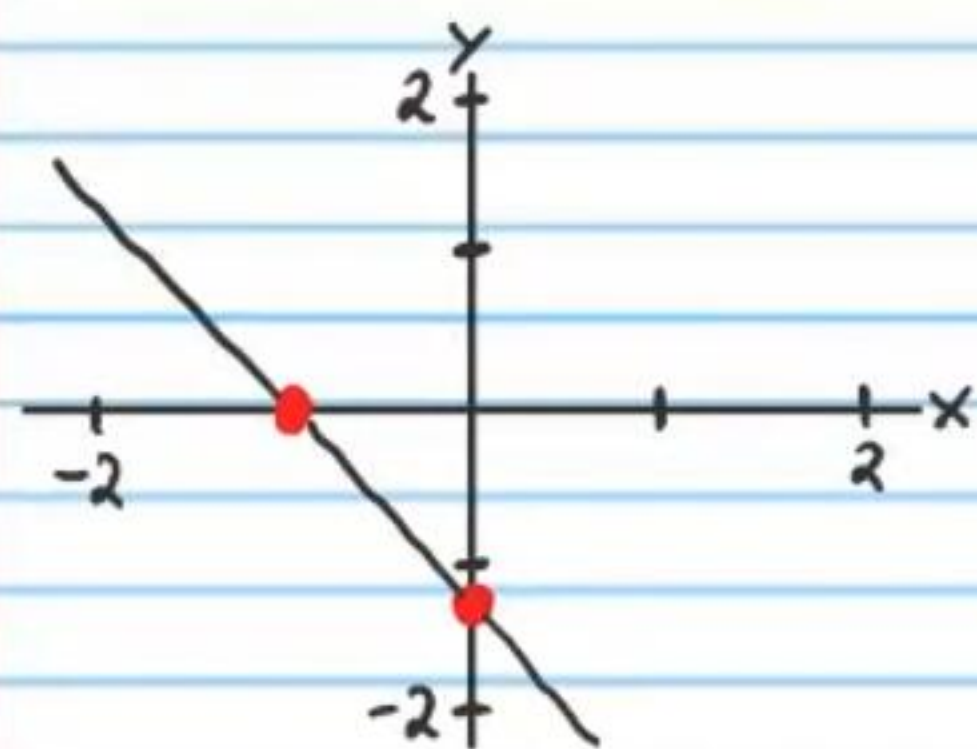
$-8 = 4Y$

$-2 = Y$

$(8, -2)$

Method II: Finding x and y Intercepts

⑬ $3y + 4x = -4$



1st Point

$x = 0$

$$\begin{aligned} \Downarrow \\ 3y + 4(0) &= -4 \\ 3y + 0 &= -4 \\ 3y &= -4 \\ y &= -\frac{4}{3} \end{aligned}$$

$(0, -\frac{4}{3})$

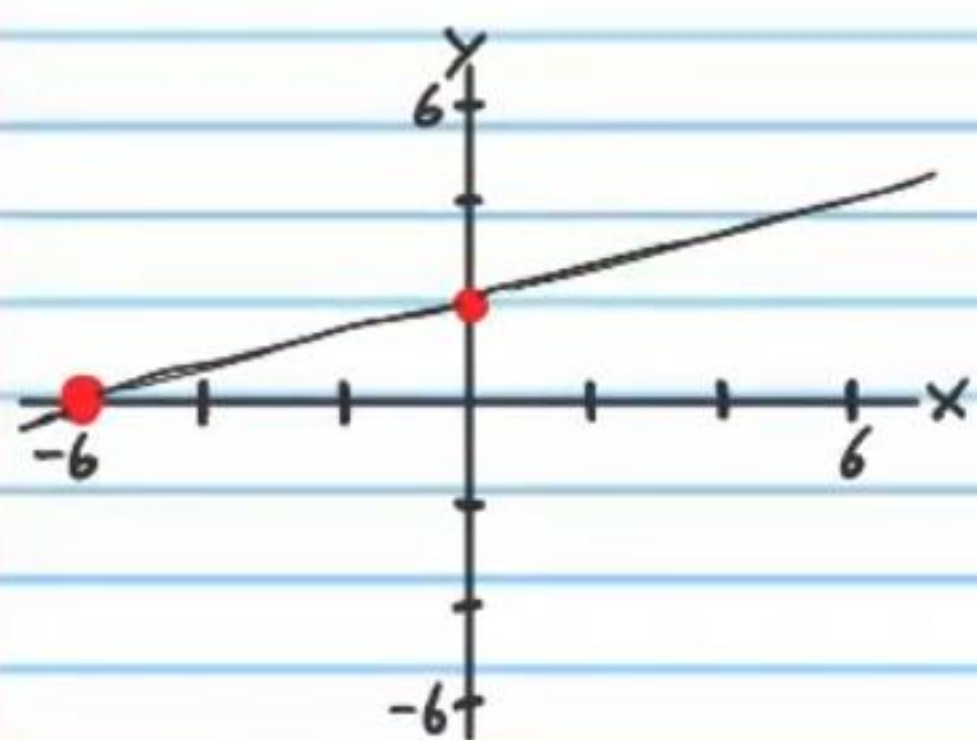
2nd Point

$y = 0$

$$\begin{aligned} \Downarrow \\ 3(0) + 4x &= -4 \\ 0 + 4x &= -4 \\ 4x &= -4 \\ x &= -1 \end{aligned}$$

$(-1, 0)$

⑭ $6 = -x + 3y$



1st Point

$y = 0$

$$\begin{aligned} \Downarrow \\ 6 &= -x + 3(0) \\ 6 &= -x + 0 \\ 6 &= -x \\ -6 &= x \end{aligned}$$

$(-6, 0)$

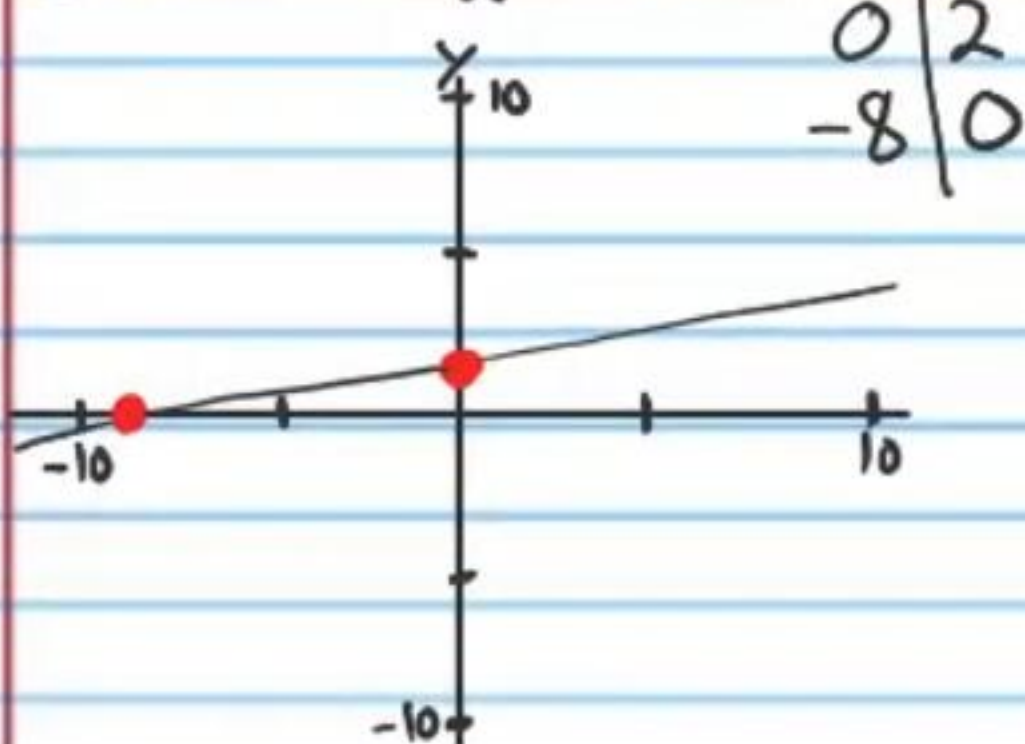
2nd Point

$x = 0$

$$\begin{aligned} \Downarrow \\ 6 &= -0 + 3y \\ 6 &= 0 + 3y \\ 6 &= 3y \\ 2 &= y \end{aligned}$$

$(0, 2)$

⑮ $-4 = \frac{x}{2} - 2y$



1st Point

$x = 0$

$$\begin{aligned} \Downarrow \\ -4 &= \frac{0}{2} - 2y \\ -4 &= 0 - 2y \\ -4 &= -2y \\ 2 &= y \end{aligned}$$

$2 = y$

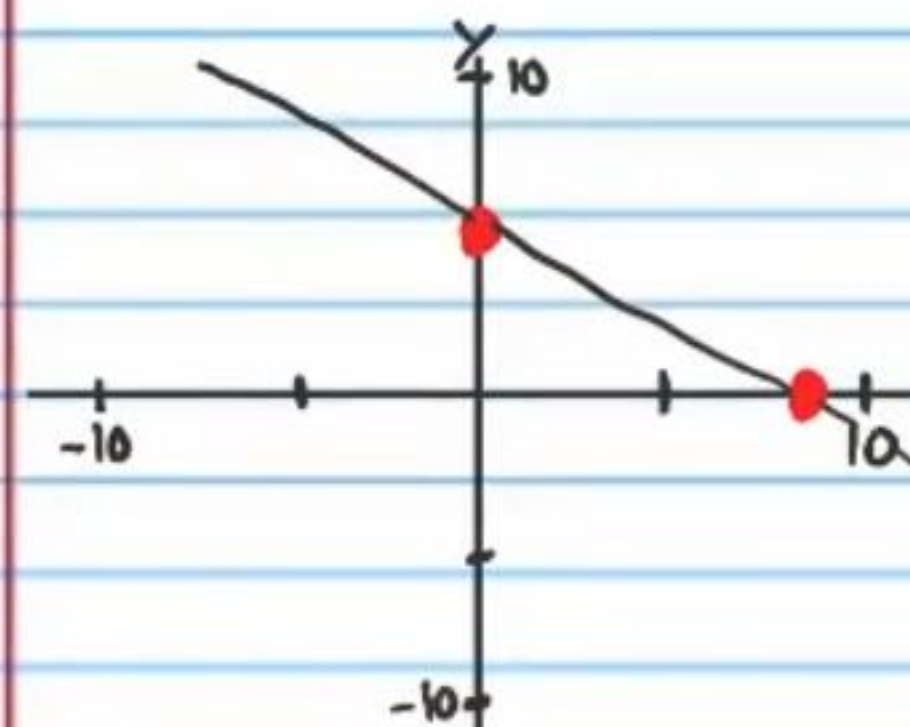
2nd Point

$y = 0$

$$\begin{aligned} \Downarrow \\ -4 &= \frac{x}{2} - 2(0) \\ -4 &= \frac{x}{2} - 0 \\ -4 &= \frac{x}{2} \\ -8 &= x \end{aligned}$$

$-8 = x$

⑯ $5x + 8y = 40$



1st Point

$x = 0$

$$\begin{aligned} \Downarrow \\ 5(0) + 8y &= 40 \\ 0 + 8y &= 40 \\ 8y &= 40 \\ y &= 5 \end{aligned}$$

$(0, 5)$

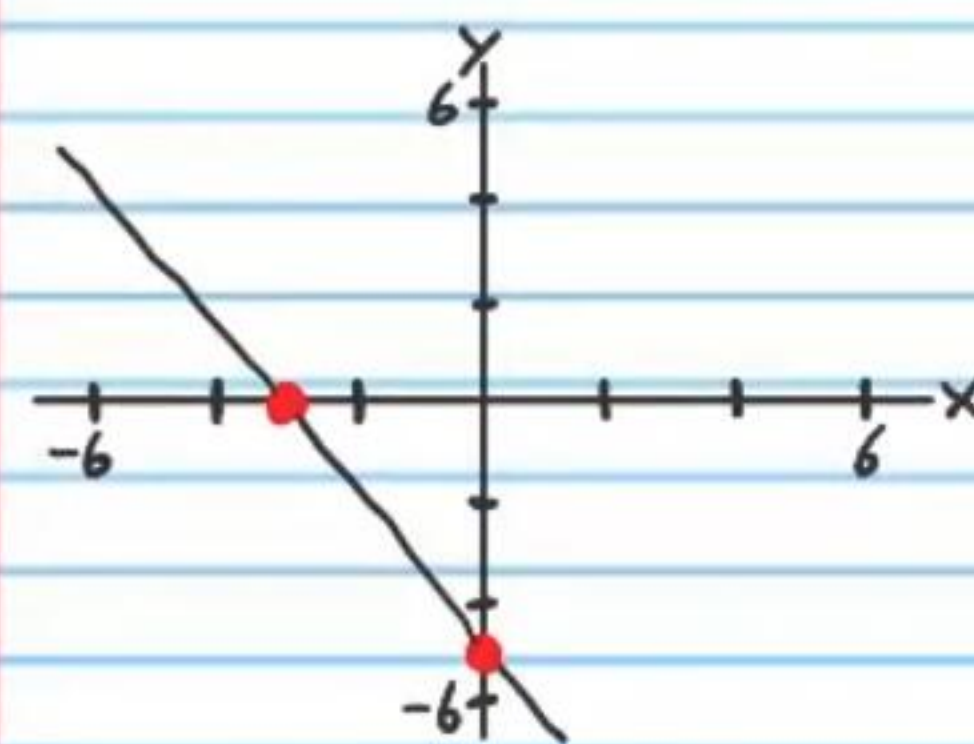
2nd Point

$y = 0$

$$\begin{aligned} \Downarrow \\ 5x + 8(0) &= 40 \\ 5x + 0 &= 40 \\ 5x &= 40 \\ x &= 8 \end{aligned}$$

$(8, 0)$

⑰ $15 = -5x - 3y$



1st Point

$x = 0$

$$\begin{aligned} \Downarrow \\ 15 &= -5(0) - 3y \\ 15 &= 0 - 3y \\ 15 &= -3y \\ -5 &= y \end{aligned}$$

$(0, -5)$

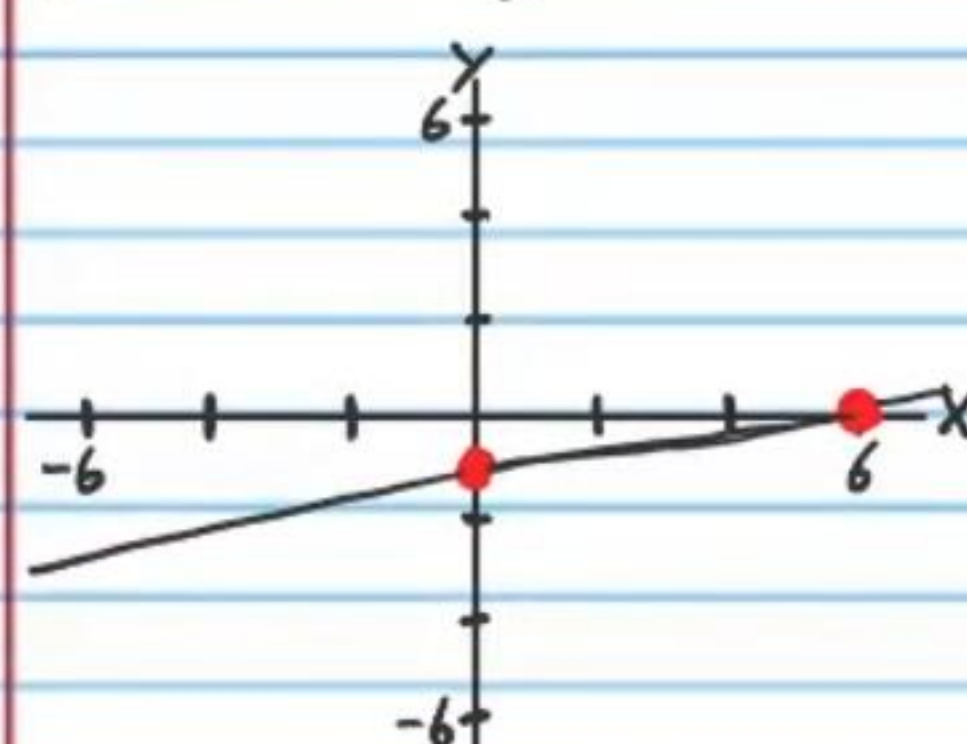
2nd Point

$y = 0$

$$\begin{aligned} \Downarrow \\ 15 &= -5x - 3(0) \\ 15 &= -5x - 0 \\ 15 &= -5x \\ -3 &= x \end{aligned}$$

$(-3, 0)$

⑱ $2 = \frac{x}{3} - 2y$



1st Point

$x = 0$

$$\begin{aligned} \Downarrow \\ 2 &= \frac{0}{3} - 2y \\ 2 &= 0 - 2y \\ 2 &= -2y \\ -1 &= y \end{aligned}$$

$(0, -1)$

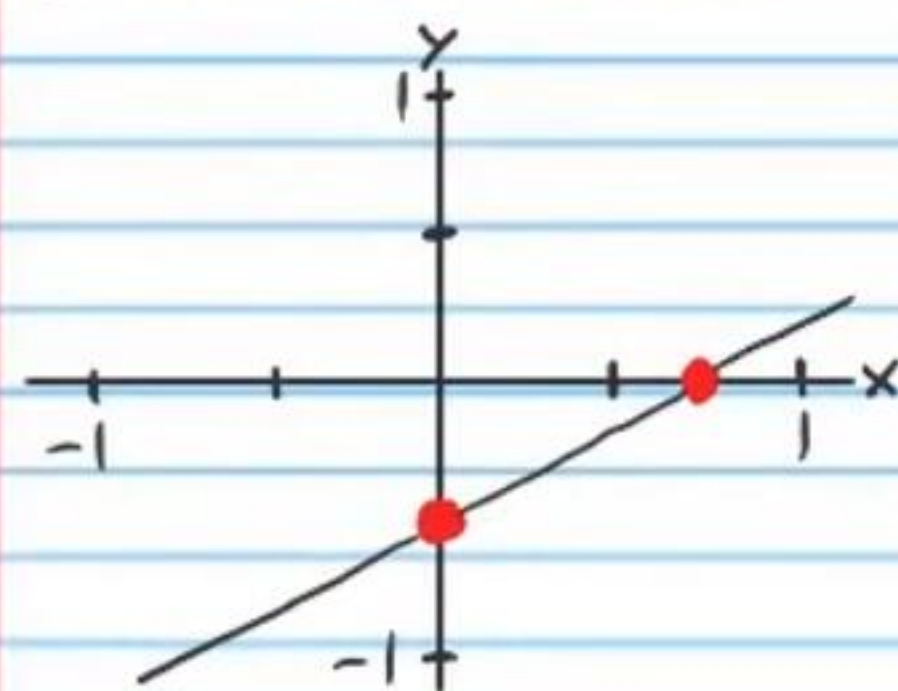
2nd Point

$y = 0$

$$\begin{aligned} \Downarrow \\ 2 &= \frac{x}{3} - 2(0) \\ 2 &= \frac{x}{3} - 0 \\ 2 &= \frac{x}{3} \\ 6 &= x \end{aligned}$$

$(6, 0)$

• (19) $3x - 4y = 2$



1st Point

$x = 0$



$$\begin{aligned} 3(0) - 4y &= 2 \\ 0 - 4y &= 2 \\ -4y &= 2 \\ y &= \frac{2}{-4} \\ y &= -\frac{1}{2} \\ (0, -\frac{1}{2}) \end{aligned}$$

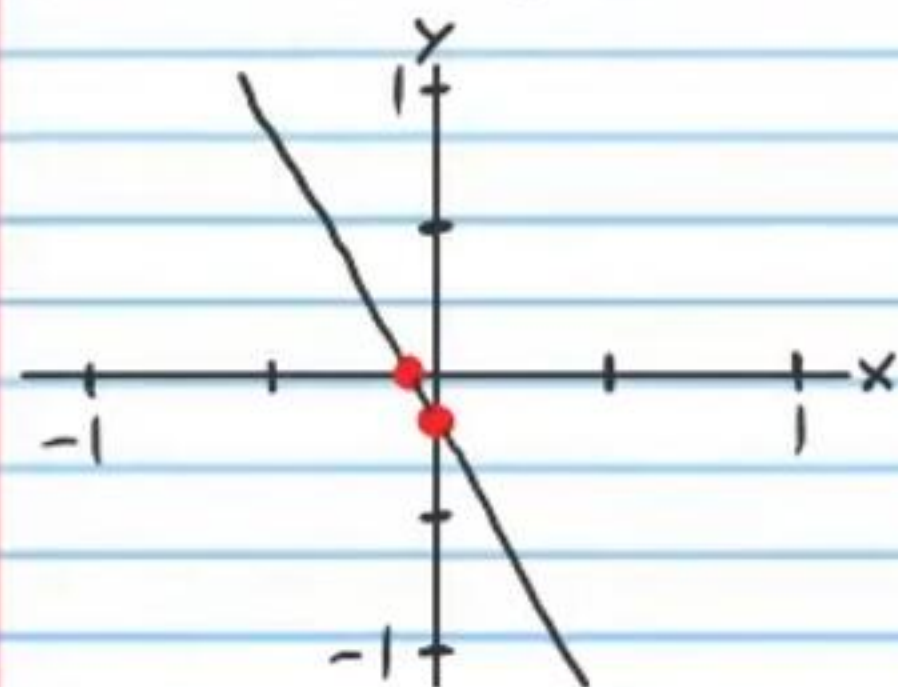
2nd Point

$y = 0$



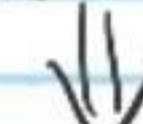
$$\begin{aligned} 3x - 4(0) &= 2 \\ 3x - 0 &= 2 \\ 3x &= 2 \\ x &= \frac{2}{3} \\ (\frac{2}{3}, 0) \end{aligned}$$

(20) $-7x - 5y = 1$



1st Point

$x = 0$



$$\begin{aligned} -7(0) - 5y &= 1 \\ 0 - 5y &= 1 \\ -5y &= 1 \\ y &= -\frac{1}{5} \\ (0, -\frac{1}{5}) \end{aligned}$$

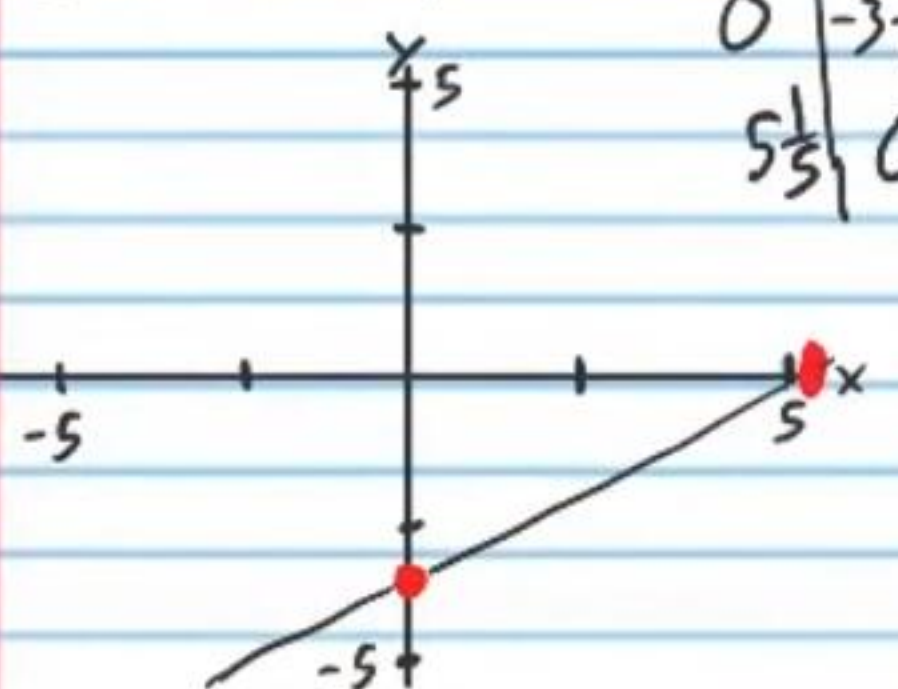
2nd Point

$y = 0$



$$\begin{aligned} -7x - 5(0) &= 1 \\ -7x - 0 &= 1 \\ -7x &= 1 \\ x &= -\frac{1}{7} \\ (-\frac{1}{7}, 0) \end{aligned}$$

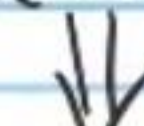
(21) $15x - 20y = 78$



x	y
0	$-3\frac{9}{10}$
$5\frac{1}{5}$	0

1st Point

$x = 0$



$$\begin{aligned} 15(0) - 20y &= 78 \\ 0 - 20y &= 78 \\ -20y &= 78 \\ y &= \frac{-78}{20} \\ y &= -\frac{39}{10} \end{aligned}$$

2nd Point

$y = 0$

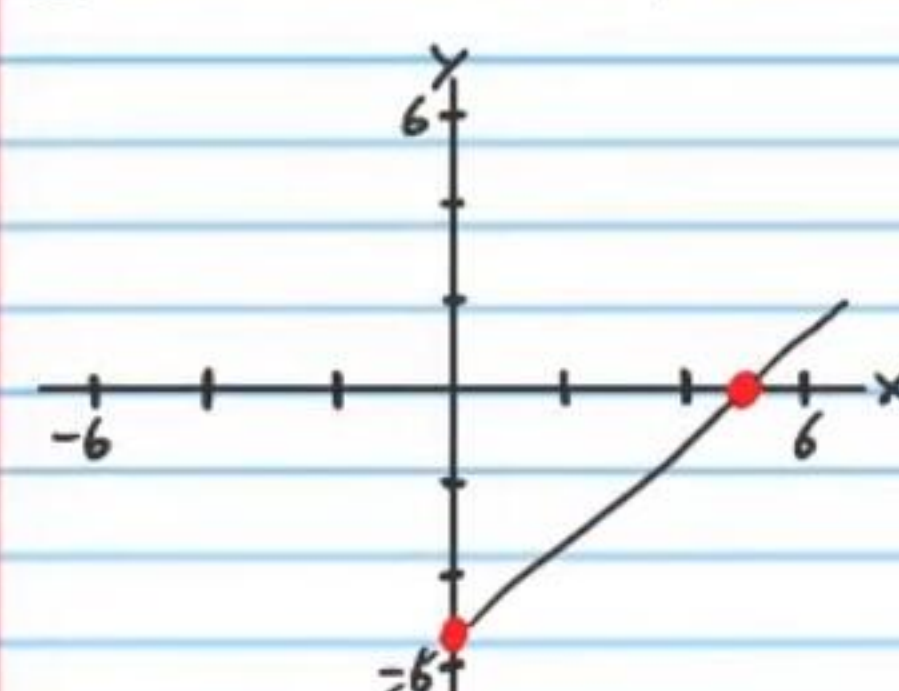


$$\begin{aligned} 15x - 20(0) &= 78 \\ 15x - 0 &= 78 \\ 15x &= 78 \\ x &= \frac{78}{15} \\ x &= 5\frac{1}{5} \end{aligned}$$

$y = -3\frac{9}{10}$
 $(0, -3\frac{9}{10})$

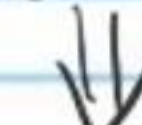
$x = 5\frac{1}{5}$
 $(5\frac{1}{5}, 0)$

(22) $40 = 8x - 7y$



1st Point

$x = 0$



$$\begin{aligned} 40 &= 8(0) - 7y \\ 40 &= 0 - 7y \\ 40 &= -7y \\ \frac{40}{-7} &= y \\ -5\frac{5}{7} &= y \\ (0, -5\frac{5}{7}) \end{aligned}$$

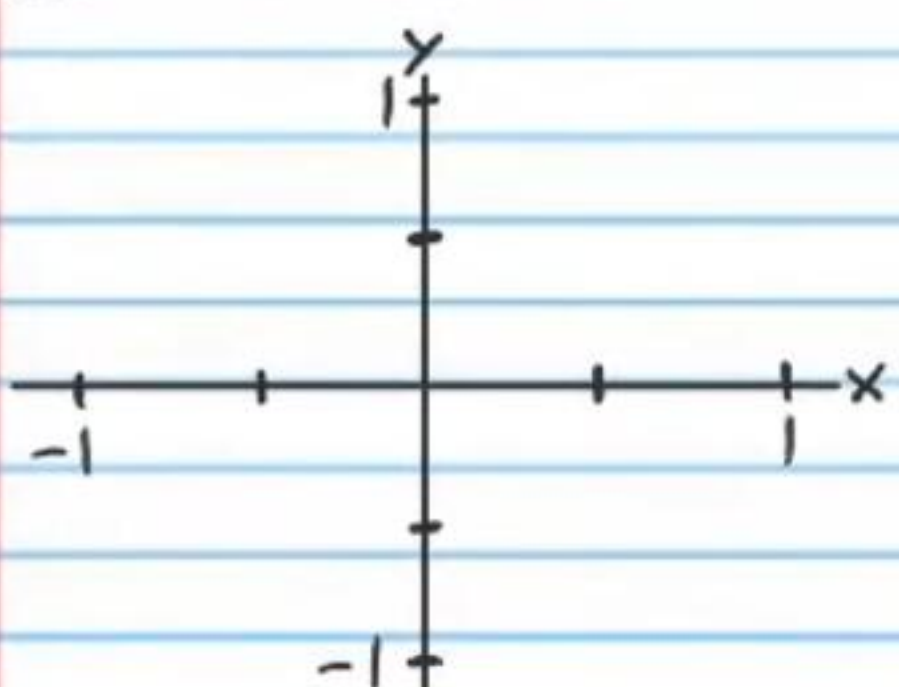
2nd Point

$y = 0$



$$\begin{aligned} 40 &= 8x - 7(0) \\ 40 &= 8x - 0 \\ 40 &= 8x \\ 5 &= x \\ (5, 0) \end{aligned}$$

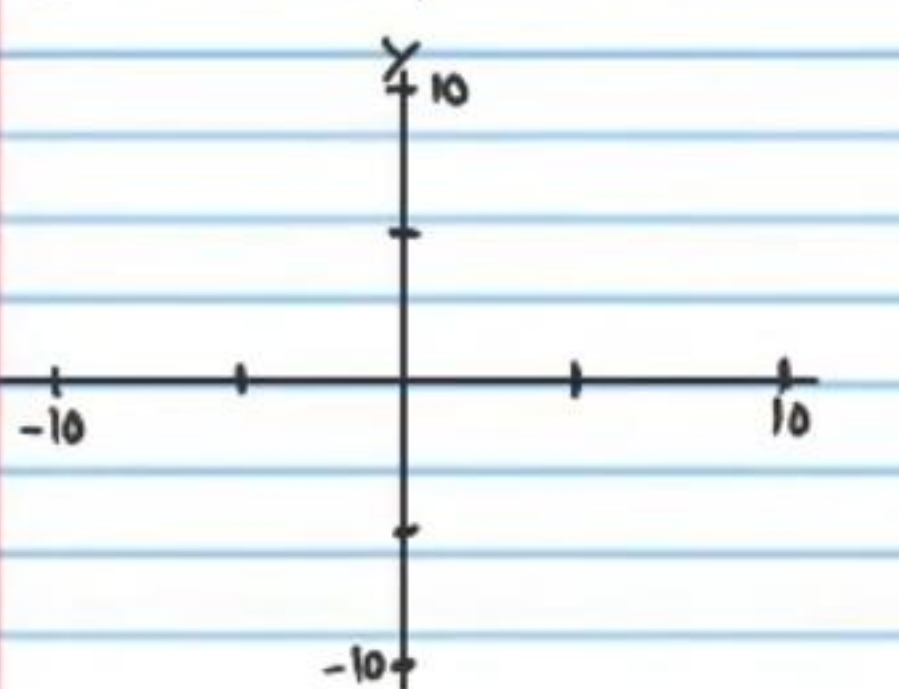
(23) $1 = -2y + 2x$



1st Point

2nd Point

(24) $21x - 20y = 100$

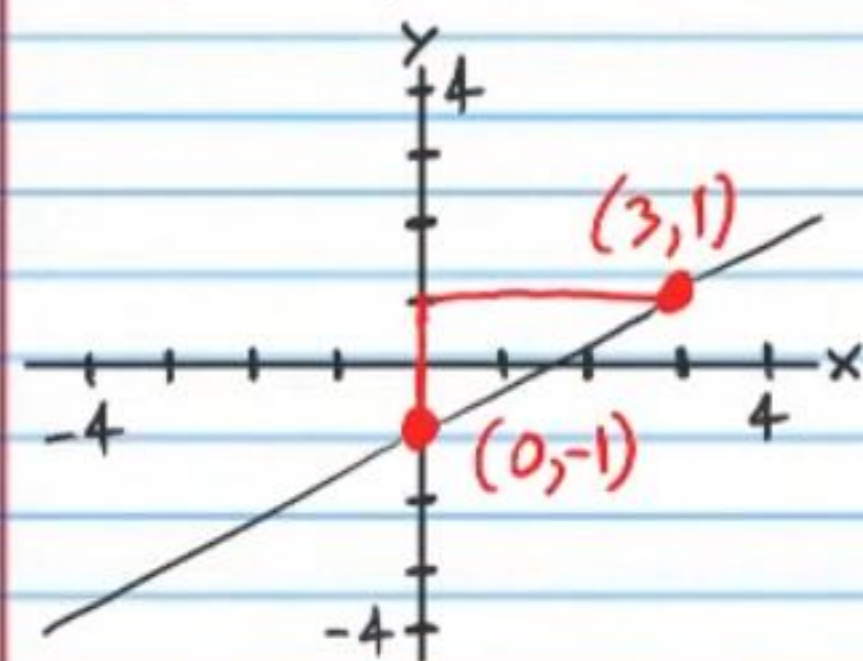


1st Point

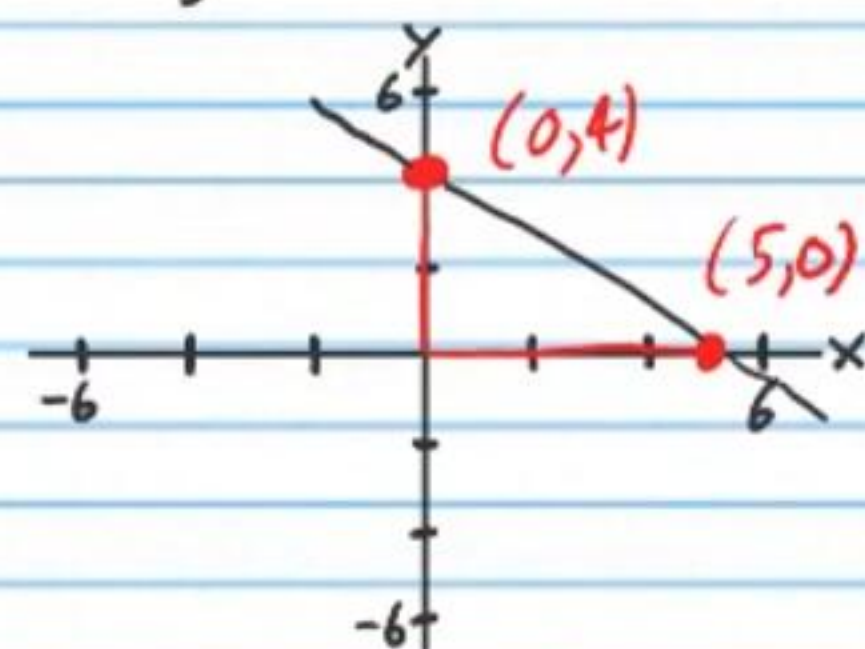
2nd Point

Method III: Convert to Slope-Intercept Form and Use Slope and Y-Intercept

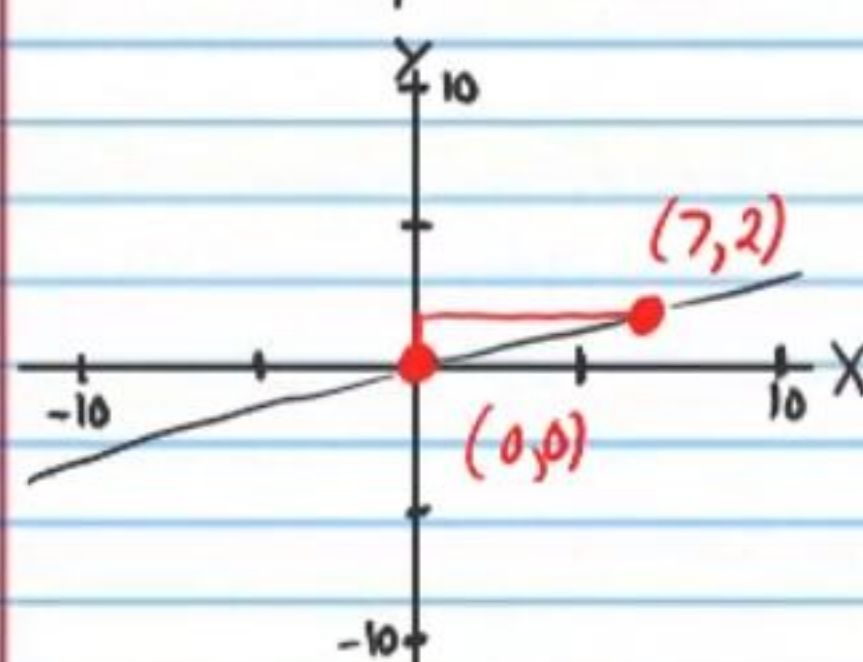
②⑤ $y = \frac{2}{3}x - 1$



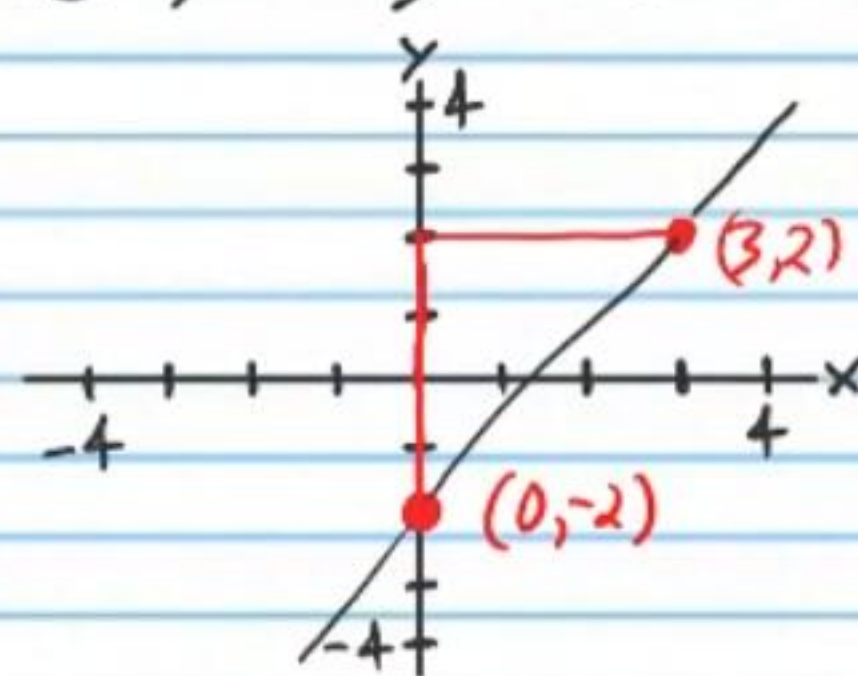
②⑥ $-\frac{4}{5}x + 4 = y$



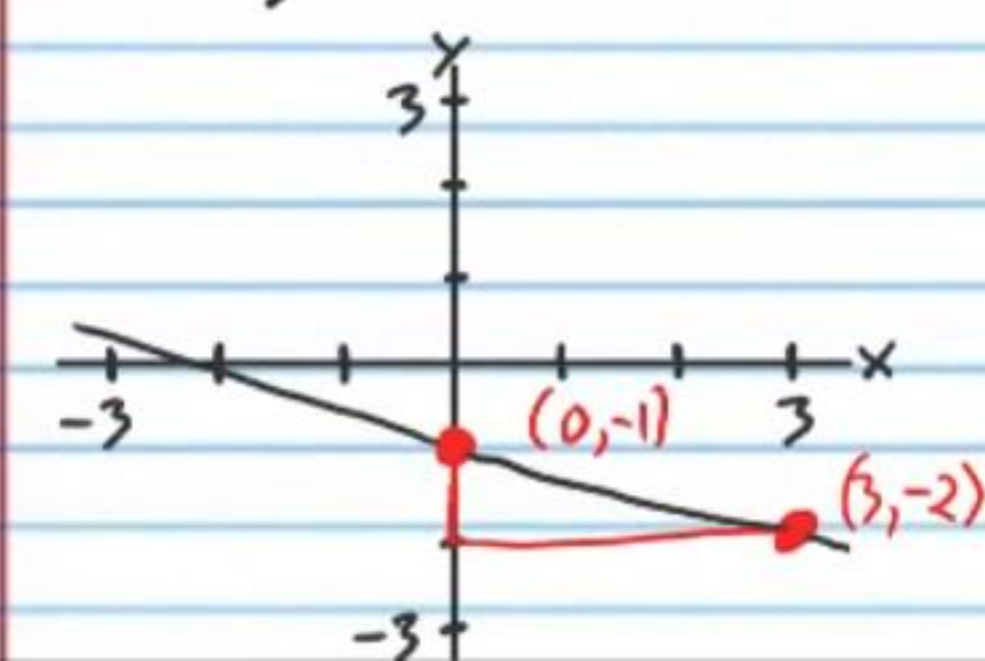
②⑦ $y = \frac{2}{7}x$



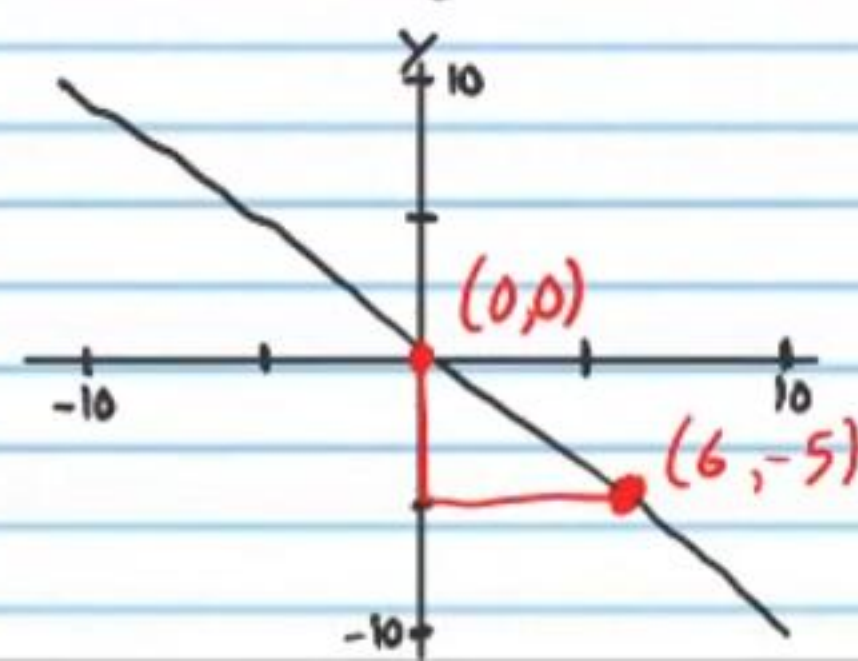
②⑧ $y = \frac{4}{3}x - 2$



②⑨ $-\frac{1}{3}x - 1 = y$

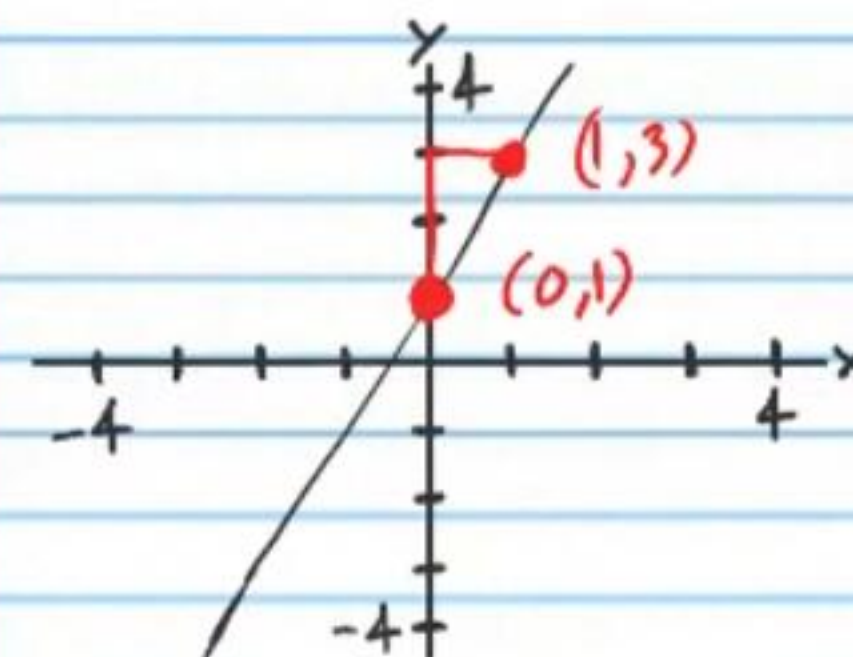


③⑩ $y = -\frac{5}{6}x$



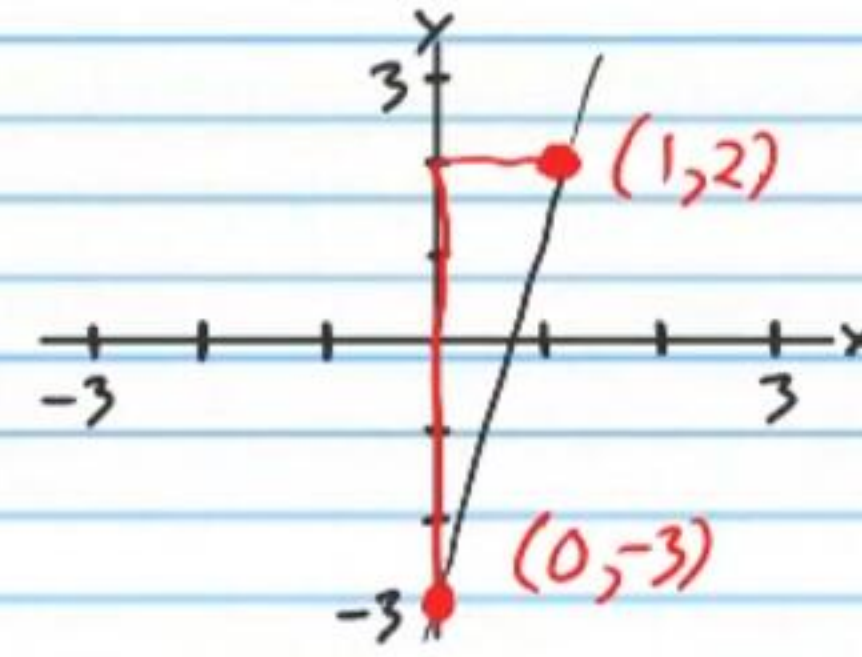
$2 = \frac{2}{1} = \frac{\text{rise}}{\text{run}}$

③① $y = 2x + 1$

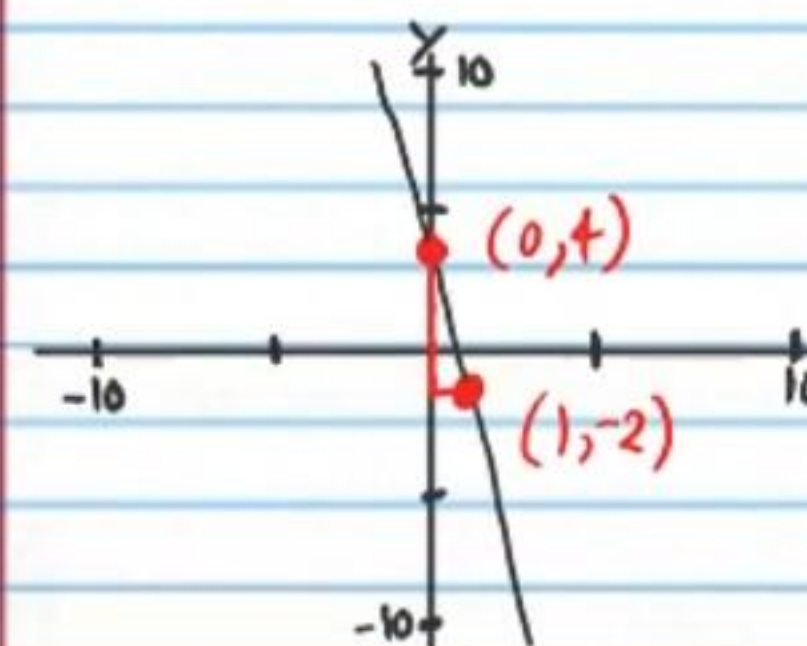


$5 = \frac{5}{1} = \frac{\text{rise}}{\text{run}}$

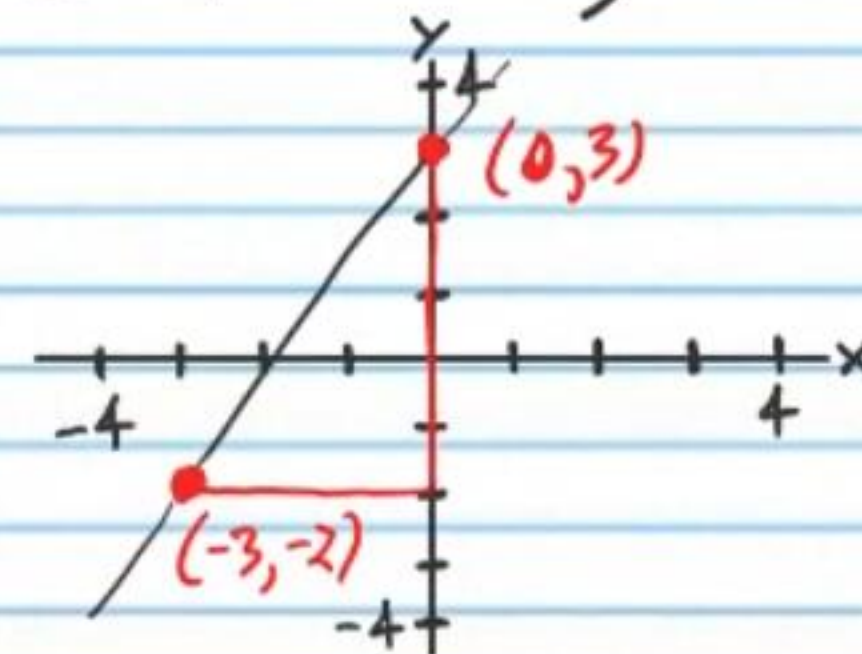
③② $5x - 3 = y$



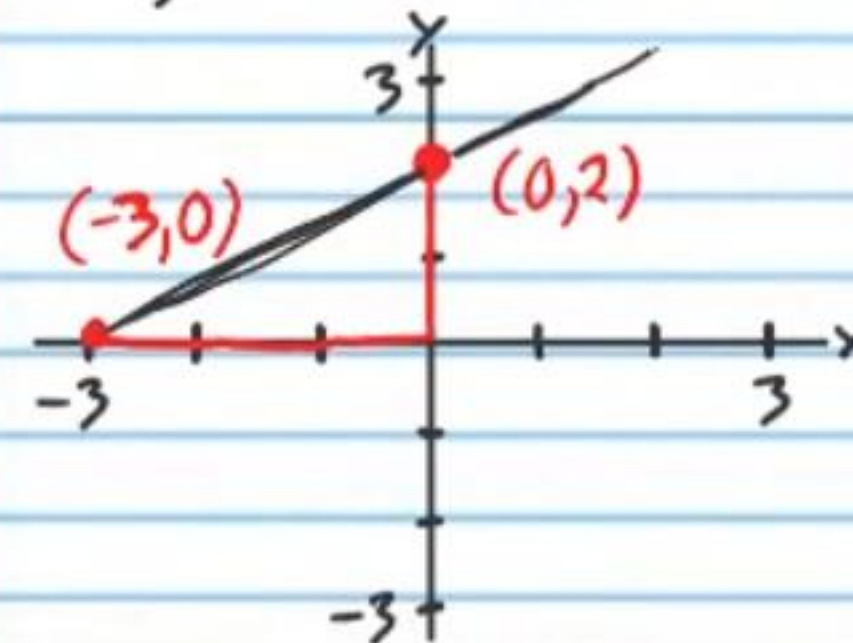
③③ $y = -6x + 4$



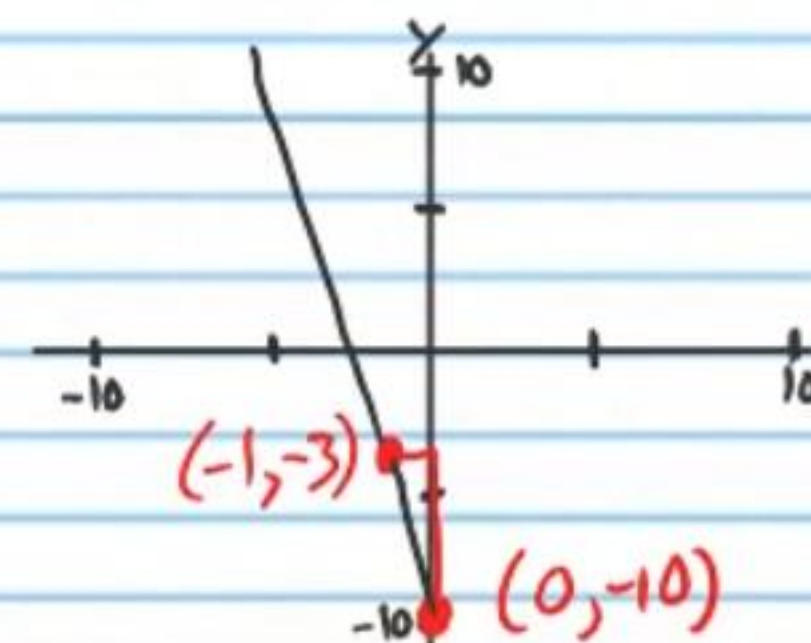
③④ $y = 3 + \frac{5}{3}x$



③⑤ $\frac{2}{3}x + 2 = y$



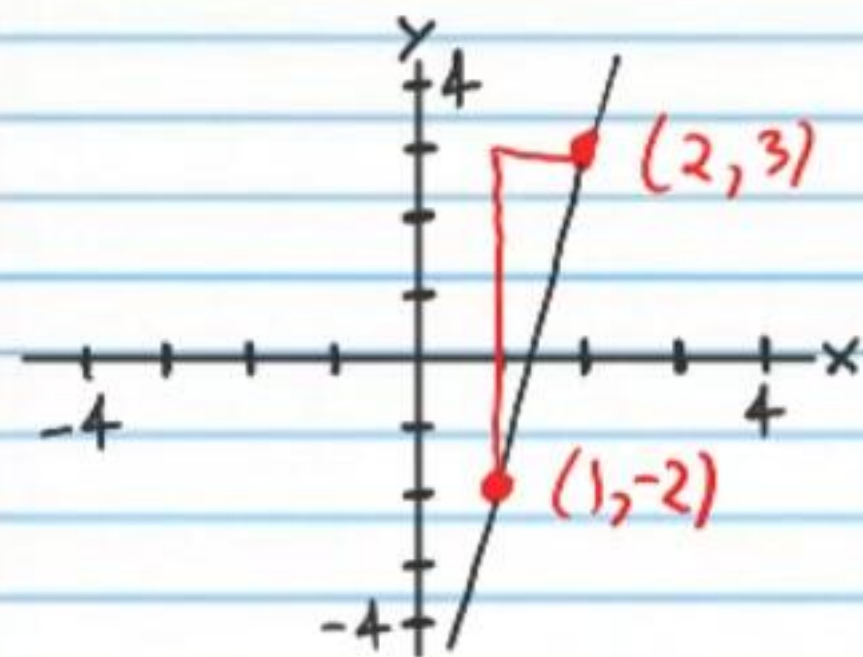
③⑥ $y = -10 - 7x$



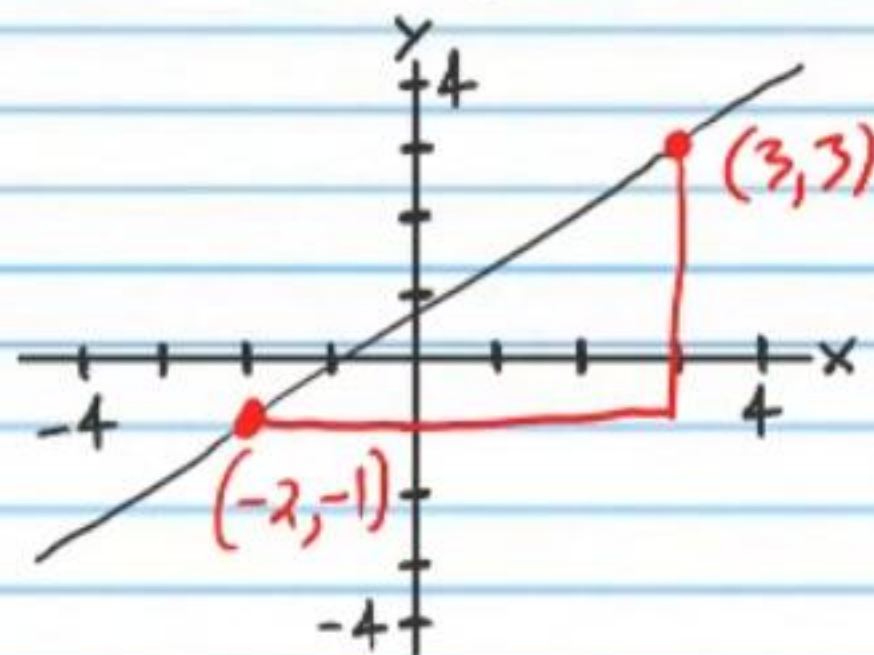
$$y - y_p = m(x - x_p)$$

Method IV: Using Point-Slope Form

③⑦ $y + 2 = 5(x - 1)$

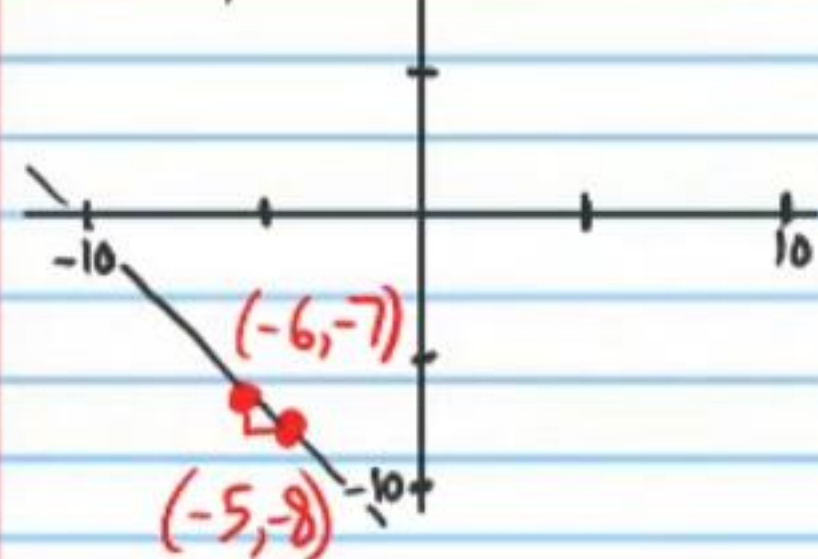


③⑧ $y - 3 = \frac{4}{5}(x - 3)$

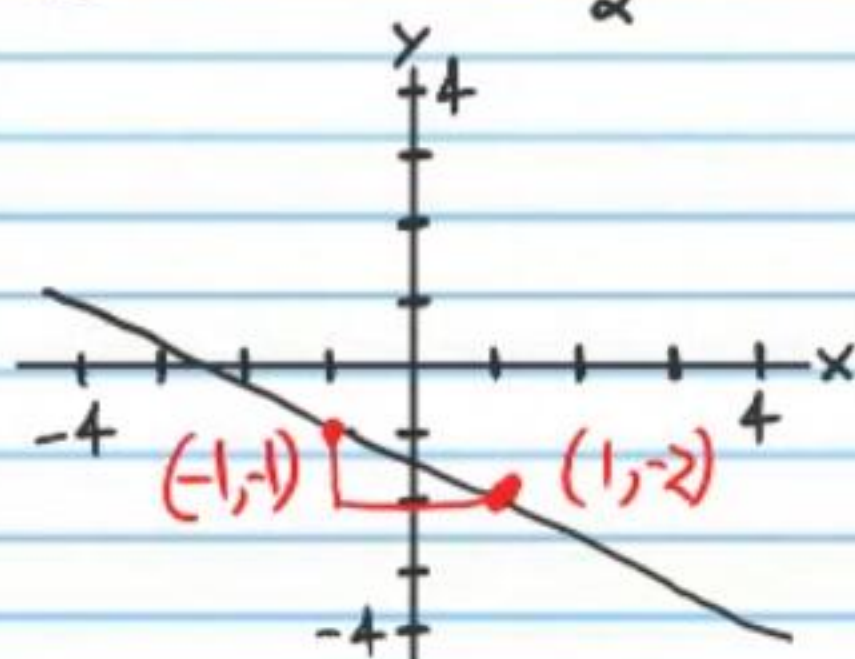


③⑨ $-(x + 6) = y + 7$

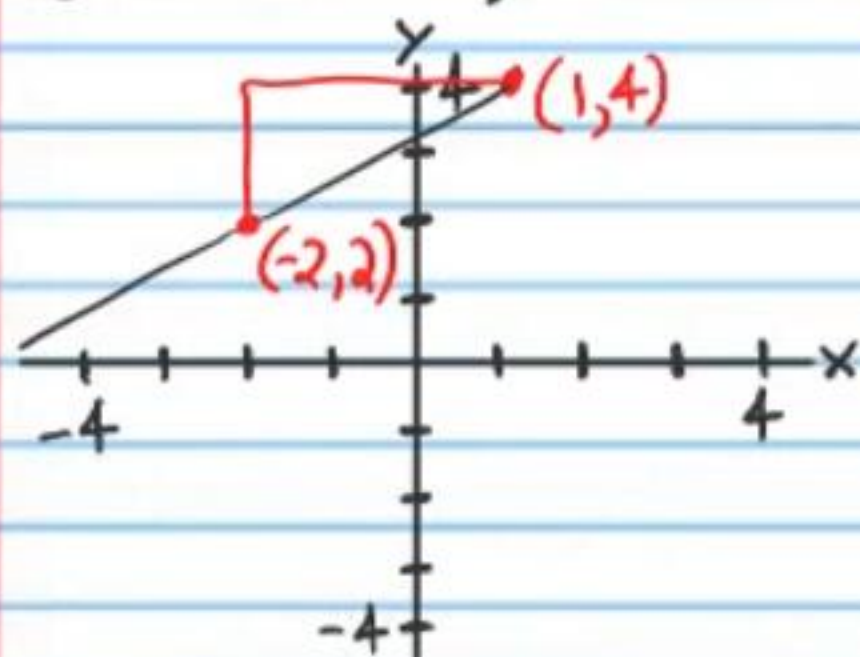
$-1 = -\frac{1}{1}$



④⑩ $y + 1 = -\frac{1}{2}(x + 1)$

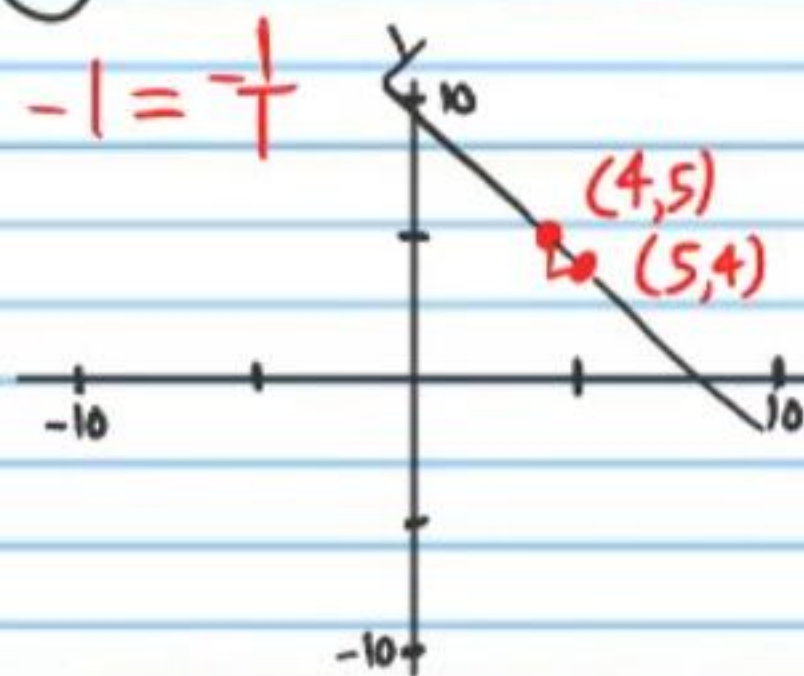


④⑪ $y - 2 = \frac{2}{3}(x + 2)$

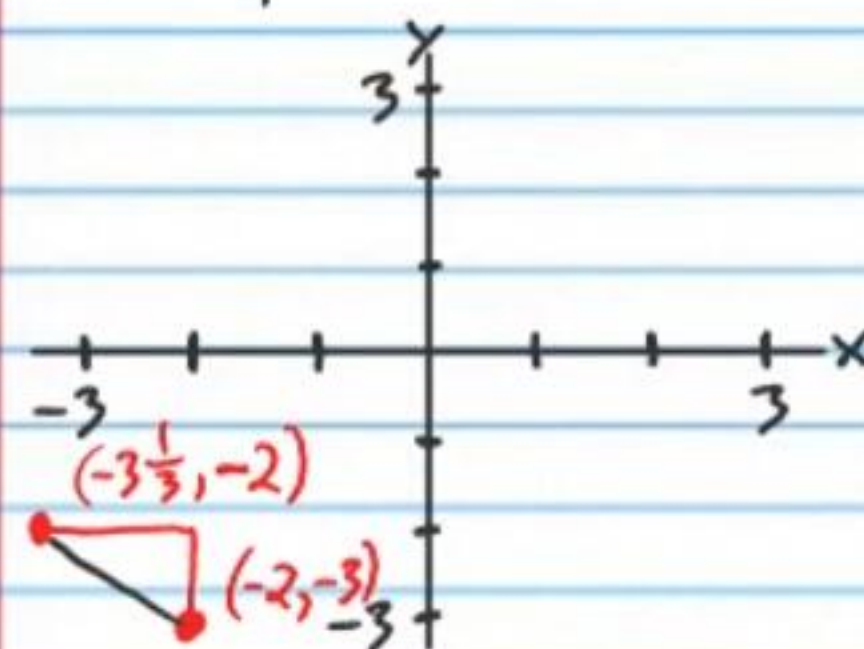


④⑫ $-(x - 4) = y - 5$

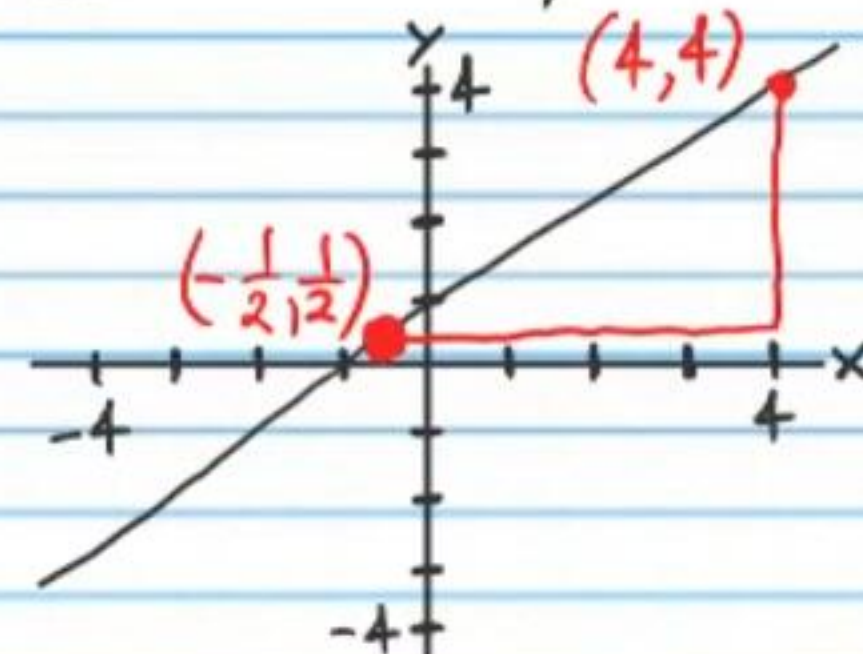
$-1 = -\frac{1}{1}$



④⑬ $-\frac{3}{4}(x + 2) = y + 3$



④⑭ $y - 4 = \frac{7}{9}(x - 4)$



④⑮ $y + 10 = -\frac{7}{15}(x - 3)$

